

Solution Guide –Adecco HR Analytics Using EXCEL

Part 1: Basic-Level Questions and Answers

Q1. What is the overall attrition rate at Adecco India?

Answer: The dataset contains 1,470 employees, of whom 237 have left the company. The overall attrition rate is approximately 16.12%.

Business insight: Approximately 16 out of every 100 employees have left the company. HR should monitor attrition regularly and identify the departments and employee groups with higher turnover.

Excel formula:

```
=COUNTIF (Attrition_Range, "Yes") / COUNTA (Employee_ID_Range)
```

Format the result as a percentage.

Q2. Which department has the highest attrition rate?

Answer: The Sales department has the highest attrition rate, at approximately 20.63%. Human Resources follows at around 19.05%, while Research & Development has a lower rate of approximately 13.84%.

Business insight: Sales employees may be experiencing challenges related to targets, workload, incentives, or career progression. HR should investigate the reasons for turnover in Sales and compare employee feedback across departments.

Excel approach: Create a PivotTable using Department and Attrition, then calculate the percentage of employees who left in each department.

Q3. What is the average age of employees who have left the company?

Answer: The average age of employees who left is approximately 33.61 years.

Business insight: This provides a demographic profile of employees who leave the company. HR can investigate career-development opportunities, compensation, and work-life balance among employees in this age group without assuming age itself causes attrition.

Excel formula:

```
=AVERAGEIF (Attrition_Range, "Yes", Age_Range)
```

Q4. How does job satisfaction vary across different job roles?

Answer: Job satisfaction varies across job roles, with scores measured on a scale from 1 to 4. A score of 1 indicates low satisfaction, while 4 indicates high satisfaction.

Business insight: Comparing average satisfaction across job roles helps HR identify teams that may need better management support, recognition, or career-development opportunities. The exact ranking of job roles should be taken from the PivotTable results in the workbook.

Excel approach: Place `JobRole` in Rows and `JobSatisfaction` in Values, then change the aggregation to Average.

Q5. Is there a significant difference in attrition rates between male and female employees?

Answer: In the dataset, the attrition rate is approximately 17.01% for male employees and 14.80% for female employees.

Business insight: Male employees have a somewhat higher attrition rate in these results. HR should investigate whether the difference is related to job roles, compensation, overtime, or other workplace factors. A statistical test is needed to determine whether the difference is significant.

Excel approach: Create a PivotTable with Gender in Rows and Attrition in Columns, then calculate the attrition percentage for each gender.

Q6. What is the average monthly income of employees who have left the company?

Answer: The average monthly income of employees who left the company is approximately 4,787.09 in the dataset's income units.

Business insight: This figure helps HR understand the compensation levels of employees who leave. Salary alone does not explain attrition, so it should also be compared with job level, department, and tenure.

Excel formula:

```
=AVERAGEIF(Attrition_Range, "Yes", MonthlyIncome_Range)
```

Q7. How does distance from home impact employee attrition?

Answer: The correlation between distance from home and attrition is approximately 0.078, indicating a very weak positive relationship.

Business insight: Employees living farther away may face additional commuting challenges, but the relationship is weak in this dataset. HR could investigate commute patterns alongside work-life balance and business travel before introducing flexible-work policies.

Q8. What is the distribution of performance ratings among employees?

Answer: The dataset contains 1,244 employees with a performance rating of 3 and 226 employees with a rating of 4.

Business insight: Most employees received a rating of 3. HR can use this distribution to understand overall performance-rating patterns and review whether employees have sufficient opportunities for training, development, and career progression.

Q9. How many employees work overtime regularly?

Answer: A total of 416 employees have `OverTime = Yes`, while 1,054 employees have `OverTime = No`. Approximately 28.30% of employees are marked as working overtime.

Business insight: Overtime may create work-life balance challenges for some employees. HR should compare attrition rates between employees who work overtime and those who do not, then review workload distribution and staffing needs.

Excel formula:

```
=COUNTIF(OverTime_Range, "Yes")
```

Q10. What is the average number of years employees have worked at Adecco India?

Answer: The average employee tenure is approximately 7.01 years.

Business insight: This provides a general view of employee retention and workforce experience. However, the average can hide differences between newer and longer-serving employees, so HR should also compare tenure groups by attrition rate.

Excel formula:

```
=AVERAGE(YearsAtCompany_Range)
```

Part 2: Medium-Level Questions and Answers

Q1. What are the top three factors contributing to employee attrition at Adecco India?

Answer: The strongest numerical relationships with attrition among the factors previously analyzed include:

- Total working years: correlation of approximately -0.171
- Job level: correlation of approximately -0.169
- Years in current role: correlation of approximately -0.161

Business insight: These weak negative relationships suggest that employees with more work experience, higher job levels, and longer time in their current roles tend to have lower attrition. However, correlation does not prove causation. HR should also examine overtime, job satisfaction, and other workplace factors.

Q2. How does the attrition rate vary with different levels of job involvement?

Answer: The reported attrition rates by job involvement level are approximately:

Job involvement level	Attrition rate
1	33.73%
2	18.93%
3	14.40%
4	9.03%

Business insight: Employees with lower job involvement have higher attrition rates in these results. HR should explore ways to improve engagement, provide meaningful responsibilities, and support employees who feel less involved in their work.

Q3. Is there a relationship between employee age and job satisfaction?

Answer: The correlation between age and job satisfaction is approximately -0.005 , indicating virtually no linear relationship.

Business insight: Age does not appear to be meaningfully associated with job satisfaction in this dataset. HR should focus on workplace factors such as management support, work-life balance, recognition, and job responsibilities rather than assuming satisfaction depends on age.

Q4. How does marital status impact employee attrition?

Answer: The reported attrition rates by marital status are:

Marital status Attrition rate

Single	25.53%
Married	12.48%
Divorced	10.09%

Business insight: Single employees have the highest attrition rate among these groups. HR can investigate whether career opportunities, compensation, or workplace conditions are associated with this pattern, without making assumptions about individual employees based on marital status.

Q5. What is the impact of training on employee attrition?

Answer: Employees with zero training sessions have a reported attrition rate of approximately 27.78%, compared with 12.68% for employees with one session.

Business insight: This suggests that training participation may be associated with employee retention. However, the pattern may vary across training levels, and the result does not prove that training directly reduces attrition. HR should evaluate training quality, career-development opportunities, and employee feedback.

Q6. How does work-life balance affect employee performance ratings?

Answer: Average performance ratings are similar across work-life balance levels, ranging from approximately 3.14 to 3.18.

Business insight: Performance ratings do not vary substantially with work-life balance in these results. HR should still support a healthy work-life balance because employee well-being matters even when performance ratings show little variation.

Q7. What is the effect of stock options on employee attrition?

Answer: The reported attrition rates by stock option level are:

Stock option level Attrition rate

0	24.41%
1	9.40%
2	7.59%
3	17.65%

Business insight: Employees with stock option levels 1 and 2 have lower attrition rates than employees with level 0. However, the pattern is not consistent across all levels. HR should examine other differences between these groups before concluding that stock options directly influence retention.

Q8. How does employee tenure relate to job satisfaction?

Answer: The correlation between years at the company and job satisfaction is approximately -0.004 , indicating virtually no linear relationship.

Business insight: Longer-serving employees do not necessarily report higher or lower job satisfaction in this dataset. HR should compare satisfaction across tenure groups and investigate other factors that influence the employee experience.

Q9. What is the impact of business travel on job satisfaction?

Answer: The reported average job satisfaction scores by business travel category are approximately:

Business travel category	Average job satisfaction
Non-travel	2.79
Frequent travel	2.79
Rare travel	2.70

Business insight: The average scores are relatively close, although employees who travel rarely have a slightly lower average score. HR should gather employee feedback and review travel requirements to determine whether travel policies need improvement.

Q10. How do years since last promotion affect employee performance?

Answer: The correlation between years since the last promotion and performance rating is approximately 0.018 , indicating a very weak linear relationship.

Business insight: The data does not show a meaningful linear association between promotion timing and performance ratings. HR should evaluate promotion opportunities, career progression, and performance outcomes separately rather than assuming that delayed promotions automatically lead to lower performance.